Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Why is it straightforward to find $f(z_k \mid x_k^{(i)})$ for the example problem in this lesson?

1 / 1 point

- ○ Because all models of RSSI have Gaussian pdfs.
- ○ Because there are at most two walls between any transmitter and the receiver.
- ○ Because the vehicle travels a specific path that is always a constant distance to the nearest wall.
- ◉ Because z_k is the sum of two quantities, where one is deterministic when $x_k^{(i)}$ is known and the other is a Gaussian random variable.

Correct
Yes, this is correct.

2. Why did I choose to approximate $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ instead of compute it exactly for the example problem in this lesson?

1 / 1 point

- ○ Because the computation required to implement $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ exactly is too high.
- ○ Because particle filters do not work when $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ is exactly Gaussian and so we must approximate $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ instead.
- ◉ Because $x_k^{(i)}$ is a complicated nonlinear function of the prior state and process noise and it may not be possible to solve for $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ exactly.
- ○ Because I wanted to demonstrate more examples of the sigma-point and coordinate-transformation methods you learned in prior lessons.

Correct
Yes, this is correct.

3. Why did I choose the specific importance density $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k) = f(x_k^{(i)} \mid x_{k-1}^{(i)})$ for the example problem in this lesson? Select all that apply.

1 / 1 point

- ☑ Because it simplifies the weight-update computation.

Correct
Yes, this is one reason that I chose this importance density.

- ☑ Because it is Gaussian and we know how to generate Gaussian random samples.

Correct
Yes, this is one reason that I chose this importance density.

- ☐ This is the only choice I could have made. $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k)$ is determined from the system model.
- ☐ Because we do not know z_k at the point in the algorithm where we need to compute $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k)$ and so we select an importance density that does not depend on z_k.